

Crossover Disambiguation in Kuramoto-Coupled Dense Associative Sequential Memory

Learning rules for resolving branch ambiguity in phase-oscillator sequence retrieval

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Outline

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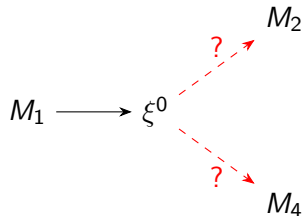
Sequential associative memory

- Hopfield memories store *static* attractors. Motor sequences, speech and episodic replay require *ordered* ones.
- Sequential retrieval = a chain of transitions $\mu \rightarrow \mu + 1$.
- **[TODO: one line: hippocampal replay + phase-native hardware]**

The crossover problem

- Two sequences share a contiguous run of patterns.
- At the shared state the network configuration is *identical* either way: the history is not encoded in the state.
- Markov-1 drives both continuations equally: the branch is chosen by noise, not by memory.

Can the coupling alone encode the route into ξ^0 ?



Objectives

- ① A Kuramoto phase-oscillator DAM for *sequences*.
- ② A learning rule that resolves crossover from the coupling alone — no external gating signal.
- ③ *When* is disambiguation possible at all, and how robust is it to corruption, network size and disorder?
- ④ **[TODO: one line: analytical / tensor contribution]**

From Hebbian storage to dense associative memory

Hopfield (1982) — pairwise Hebbian

$$J_{ij} = \frac{1}{N} \sum_{\mu} \xi_i^{\mu} \xi_j^{\mu}:$$

$$E = -\frac{N}{2} \sum_{\mu} (m^{\mu})^2, \quad m^{\mu} = \frac{1}{N} \sum_j \xi_j^{\mu} s_j$$

Crosstalk caps capacity at $\approx 0.14N$.

Krotov–Hopfield DAM (2016) — sharpen the nonlinearity, i.e. higher-order interactions:

$$E = - \sum_{\mu} F(m^{\mu}), \quad F(x) = x^{d+1}$$

Capacity $\mathcal{O}(N^d)$; minima sharpen with d .

Resulting drive

Descending E gives the *pure-power* drive $\sum_{\mu} \xi_i^{\mu} (m^{\mu})^d$ — one pattern index per term, exponent d as in Krotov DAM. Every rule that follows changes only *which* pattern each factor reads.

From the DAM energy to the Kuramoto sequence drive

- 1 **Spins $\rightarrow$ phases:** $s_i = \cos \theta_i$, so $\dot{\theta}_i = -\partial_{\theta_i} E = \omega_i - \sin \theta_i \sum_{\mu} \xi_i^{\mu} (m^{\mu})^d$.
- 2 **Symmetric ($\mu \rightarrow \mu$):** true gradient flow; stored patterns are minima.
- 3 **Asymmetric ($\mu \rightarrow \mu + 1$):** drive the *successor* (Sompolinsky–Kanter): stored patterns cease to be resting points and the state traverses the chain.

Absence of an energy functional

Asymmetric couplings $\Rightarrow$ no potential with $\dot{\theta} = -\nabla E$. “Minima”, “saddles”, “symmetry of the landscape” below refer to a **Lyapunov function** of the flow. The absence of gradient structure is precisely what permits transitions.

- **Hopfield (1982)** — pairwise Hebbian attractor memory.
- **Krotov & Hopfield (2016)** — dense associative memory: power- d interactions, super-linear capacity.
- **Sompolinsky & Kanter (1986)** — asymmetric $\mu \rightarrow \mu + 1$ couplings turn attractors into a sequence.
- **Kuramoto (1975)** — the phase-oscillator substrate.
- **Nägerl & Berloff (2025)**, arXiv:2507.21984 — DAM in a higher-order Kuramoto network; the base model for this work.
- **[TODO: oscillator / photonic hardware refs; Markov-chain sequence refs]**

Open problem: crossover disambiguation

No learning rule for crossover disambiguation

Existing formalisms specify *how* one pattern drives the next, but not what happens when two sequences pass through the *same* state.

- A transition matrix only ever sums pure powers, $\sum_{\beta} T_{\alpha\beta}(m^{\beta})^d$.
- It cannot represent a conjunction: *drive $\xi^{\mu+1}$ only if m_{μ} **and** $m_{\mu-1}$ are simultaneously large.*
- $\Rightarrow$ gate on a *product of overlaps at different lags.*

Motivation for Kuramoto dynamics

- **Smooth retrieval.** Continuous phases resolve the traversal of a crossover *in time*, rather than as a discrete jump.
- **Physical realizability.** Rules expressed purely as oscillator coupling are natively executable on analog and photonic hardware.
- **[TODO: theta–gamma phase coding, if used in Applications]**

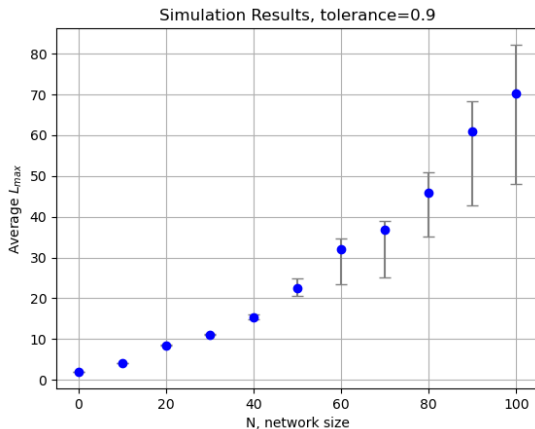
Baseline rule and retrieval algorithm

$$\dot{\theta}_i = \omega_i - \sin \theta_i \sum_{\mu} \xi_i^{\mu+1} (m_i^{\mu})^d, \quad m_i^{\mu} = \frac{1}{N-1} \sum_{j \neq i} \xi_j^{\mu} \cos \theta_j$$

Read the current overlap, drive the next pattern: Sompolinsky–Kanter at Krotov power d .

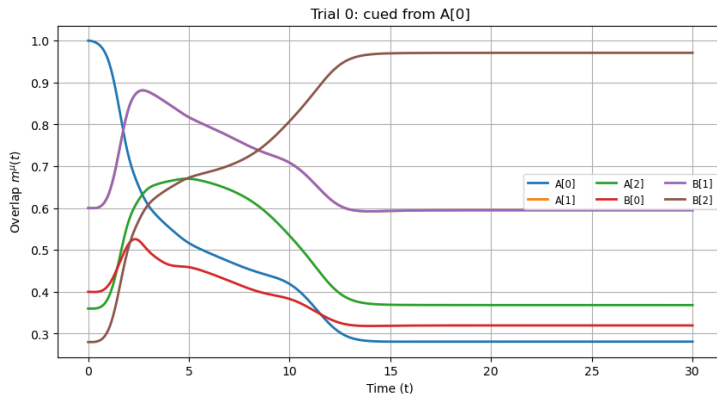
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- 1: cue $\theta(0) \leftarrow$ (possibly corrupted) ξ^0
 - 2: **for** each step **do**
 - 3: m_i^{μ} for all μ ; $V_i = \sum_{\mu} \xi_i^{\mu+1} (m_i^{\mu})^d$
 - 4: $\theta_i += dt (\omega_i - \sin \theta_i V_i)$ (+ noise)
 - 5: **end for**
 - 6: decode the sequence from overlap peaks
-

Single-sequence retrieval, baseline rule



[TODO: caption: clean peak-to-peak retrieval through P patterns]

Baseline rule at a crossover



Both continuations rise together out of the shared state. Branch choice at chance: **[TODO: XX%]** vs. 50%.

Innate symmetry of the Lyapunov function

- Couplings are asymmetric $\Rightarrow$ no energy; the flow is organized by a Lyapunov function V built from the same pure powers.
 - Every term of V reads *one* pattern index — nothing in it records *how* the state reached the shared pattern.
 - At ξ^* the gate depends only on $m(\xi^*)$, identical for both branches: the two exits are related by an exact symmetry of V (a degenerate saddle).
- $\Rightarrow$ The symmetry must be broken by information the current state cannot carry: overlaps at *several lags*, multiplied rather than summed within a single term.

Heterogeneous lag-context rules

$$\dot{\theta}_i = \omega_i - \sin \theta_i \sum_{\mu} \sum_{\text{terms}} w \xi_i^{\mu+s} \prod_{l \in \text{lags}} m_i^{\mu-l}$$

- A rule is a *table of terms*: target offset s , d source lags, weight w . Lags $(0, 0, 0)$ recovers the forward rule.
- Different lags in one term $\Rightarrow$ the gate encodes *which branch* produced the current state.

Tag	Geometry	P	Shared run L	Reach needed
2-1-2	one shared pattern	5	1	1
2-2-2	shared segment	6	2	2
2-3-2	failure case	7	3	3 (beyond catalogue)

- **Scoring:** every *unshared* pattern of the cued branch decoded, in order. Shared positions carry no rule information and are excluded.
- **Branch diagnostic:** correct vs. incorrect drive at the exit; separates incorrect selection from correct selection without lock-on.

Catalogue of learning rules

Rule	Gate at pattern μ	Reach
self	$\xi^\mu m_\mu^3$	0 (control)
forward (baseline)	$\xi^{\mu+1} m_\mu^3$	0
skip-two	$\xi^{\mu+2} m_\mu^3$	0
bigram	$\xi^{\mu+1} m_\mu m_{\mu-1} ^2$	1
multilag	$\xi^{\mu+1} m_\mu \sum_k w_k m_{\mu-k} ^2$	$\leq K$
coh bigram	$\xi^{\mu+1} m_\mu^2 m_{\mu-1}$	1
coh trigram	$\xi^{\mu+1} m_\mu m_{\mu-1} m_{\mu-2}$	2

Reach $\geq L$ is required to access *any* unshared information: necessary, not sufficient.

Experimental conditions

- $d = 3$; corruption ratio ρ sets consecutive-pattern overlap to exactly $1 - 2\rho$.
- $\omega_i \sim \mathcal{N}(\mu_\omega, \sigma_\omega^2)$ — required to break the exact-cue degeneracy.
- Euler / Euler–Maruyama, step dt , stationary early-stop.
- **Proportional gain:** rescale each term by $(1 - 2\rho)^{-\sum \text{lags}}$ to undo the geometric decay of lagged overlaps. Both raw and gain-corrected catalogues are evaluated.
- **[TODO: values: N , P , ρ , dt , T , trials, tolerance]**

Crossover resolution

- forward: at chance on branch choice; it commits rapidly, to the incorrect branch in half of trials.
- Context rules: well above chance, but frequently trade completeness for it — lagged factors leave fewer for self-amplification.
- coh bigram keeps two factors on the current overlap ($m_\mu^2 m_{\mu-1}$): it both selects the correct branch and commits to it.
- **[TODO: outcome + branch-choice bar charts, per geometry]**

Superposition of rules

- Mixing two context rules at weight w trades commitment against branch information continuously.
- `multilag` (w_0, w_1, w_2): w_0 controls commitment, w_1, w_2 supply branch information.
- **[TODO: mixture sweep grid]**

Robustness: corruption ratio and network size

$N=0$, $\text{avg_lmax}=2.0$

$N=50$, $\text{avg_lmax}=28.1$

$N=100$, $\text{avg_lmax}=159.0$

$N=150$, $\text{avg_lmax}=1273.05$

[TODO: caption: retrieval rate vs. ρ and vs. N , per rule; breakdown point = largest ρN with rate $\geq 50\%$]

Findings I: learning rules

- **Reach $\geq L$ is necessary, not sufficient.** The discriminating signal is a *dynamic residual* between traversed and untraversed predecessor; it decays within ~ 2 steps.
- **Commitment against branch information** is the governing trade-off; **[TODO: best rule]** achieves the best balance.
- **Gain correction is not uniformly beneficial:** it recovers the hardest geometry while destabilizing rules that already succeeded.
- **No rule resolves $L \gtrsim 3$,** and uncorrelated random patterns give 0% for *every* rule: sequential similarity structure is a prerequisite for all rules tested.

Findings II: dynamics and scaling

- **U-shaped dependence on the corruption ratio.** Retrieval peaks at intermediate ρ and degrades in *both* directions.
 - *Small* Hamming distance: patterns are near-duplicates and their overlaps differ by thousandths. In such a tightly packed Lyapunov landscape the rules cannot separate them.
 - *Large* ρ : consecutive overlaps are too small to sustain the cascade.
- **Frequency disorder and noise are necessary.** At $\omega \equiv 0$ an exact cue satisfies $\sin \theta_i = 0$ for all i : the drive vanishes identically and the system stalls at the cue.
- **Network size and frequency disorder must be scaled jointly.** Larger N stores more patterns with less crosstalk, but as the overlaps concentrate a fixed σ_ω crosses the rules' disorder tolerance and retrieval collapses. Two equivalent remedies: reduce σ_ω as N grows, or hold it and raise the drive with the proportional-gain filter.

Tensor structure: symmetric vs. asymmetric

- [TODO: write the coupling as $J_i^{\mu_1 \dots \mu_d, \nu}$ and split it under permutation of the read indices]
- [TODO: forward rule = fully symmetric (pure-power) part; lag-context rules = mixed-symmetry part]
- [TODO: what the decomposition yields: stability conditions, or a clean statement of the reach requirement]

- [TODO: cost per step: $\mathcal{O}(KPN)$ buffer + $\mathcal{O}(\text{terms} \times N)$ drive]
- [TODO: capacity in N, d — do context terms cost capacity?]
- [TODO: Jona: scaling law / bound]

Main result

To resolve branching after a shared segment, the learning rule needs a **lag reference exceeding the length of the shared segment**. Only a factor reading back past the last shared pattern breaks the innate symmetry of the Lyapunov function; rules whose lags all sit inside the shared run are symmetric in the branches by construction.

- Markov-1 Kuramoto DAM retrieves single sequences reliably, but its Lyapunov function is symmetric at every shared state, leaving the branch choice to noise.
- Sufficient reach is necessary, not sufficient: commitment and branch information trade off.
- The operating point is set by ρ , N and σ_ω *together*, not by the rule alone.

- **Physical simulators.** Every rule is pure oscillator coupling, and therefore maps onto analog and photonic Kuramoto hardware: crossover-resolving memory as a physical primitive.
- **Neurobiology.** [TODO: phase-coded replay, theta–gamma; lag-context gating as a disambiguation mechanism]
- [TODO: motor-sequence chaining, time-series disambiguation]

- ① How are these learning rules implemented in **physical (oscillator-based) simulators**?
- ② What is the role of **decaying neurons**, and can they replace the explicit lag buffer?
- ③ How does one generate **robust rules for randomly generated pattern sequences**?

Further open questions: whether the resolvable shared-run length L admits a fundamental ceiling, and whether rule weights can be *learned* rather than constructed by hand.

Question 1: physical implementation

- Rules are rank- $(d+1)$ couplings; hardware (Ising machines, OPO and polariton networks, CMOS oscillatory NNs) gives *pairwise*. Which mechanism synthesizes the product: ancilla modes, parametric mixing, or measurement feedback?
- Can the lagged factor $m_{\mu-\ell}$ be a physical *delay line* rather than a stored buffer? This would render the rule local in time as well as in space.
- **[TODO: required coupling precision and noise floor before branch choice falls to chance]**

Question 2: decaying neurons

- Replace the discrete lag by a decaying trace $\tau \dot{h}_\mu = -h_\mu + |m_\mu|^2$. Adaptation achieves the same from the opposite direction: the active pattern suppresses itself and releases the flow to its successor.
- Open: whether a single time constant τ provides reach beyond the shared segment, and whether it is simpler to realize physically than a multi-lag product.

Selected references

Delayed / slow synapses: Sompolinsky & Kanter (PRL 57, 2861, 1986); Kleinfeld (PNAS 83, 9469, 1986); Tank & Hopfield (PNAS 84, 1896, 1987). **Adaptation-driven transitions:** Horn & Usher (Phys. Rev. A 40, 1036, 1989); Herrmann, Ruppin & Usher (Biol. Cybern. 68, 455, 1993); Treves (Cogn. Neuropsych. 22, 276, 2005) — latching; Recanatesi et al. (Front. Comput. Neurosci. 9:149, 2015); Gillett, Pereira & Brunel (PNAS 117, 29948, 2020). **Closest precedent:** Martinez Mayorquin et al. (ICANN 2019, pp. 793–805) — filtered synaptic traces for *sequence disambiguation*.

Question 3: randomly generated sequences

- Uncorrelated patterns: 0% retrieval for every rule, raw and gain-corrected — peak overlaps 0.16–0.19 against a 0.5 tolerance.
- Gating failure, not drive failure: for independent patterns $m_{\mu-\ell} = \mathcal{O}(N^{-1/2})$; rescaling a gate that reads $m \approx 0$ has no effect.
- To test: whitened / pseudo-inverse storage (Chaudhry et al., *Long Sequence Hopfield Memory*); larger d ; ridge-fitted weights as an upper-bound control.
- **[TODO: retrieval rate vs. pattern correlation, mutation chain $\rightarrow$ fully random: locate the collapse threshold per rule]**

Thank you

[TODO: contact / repo QR]